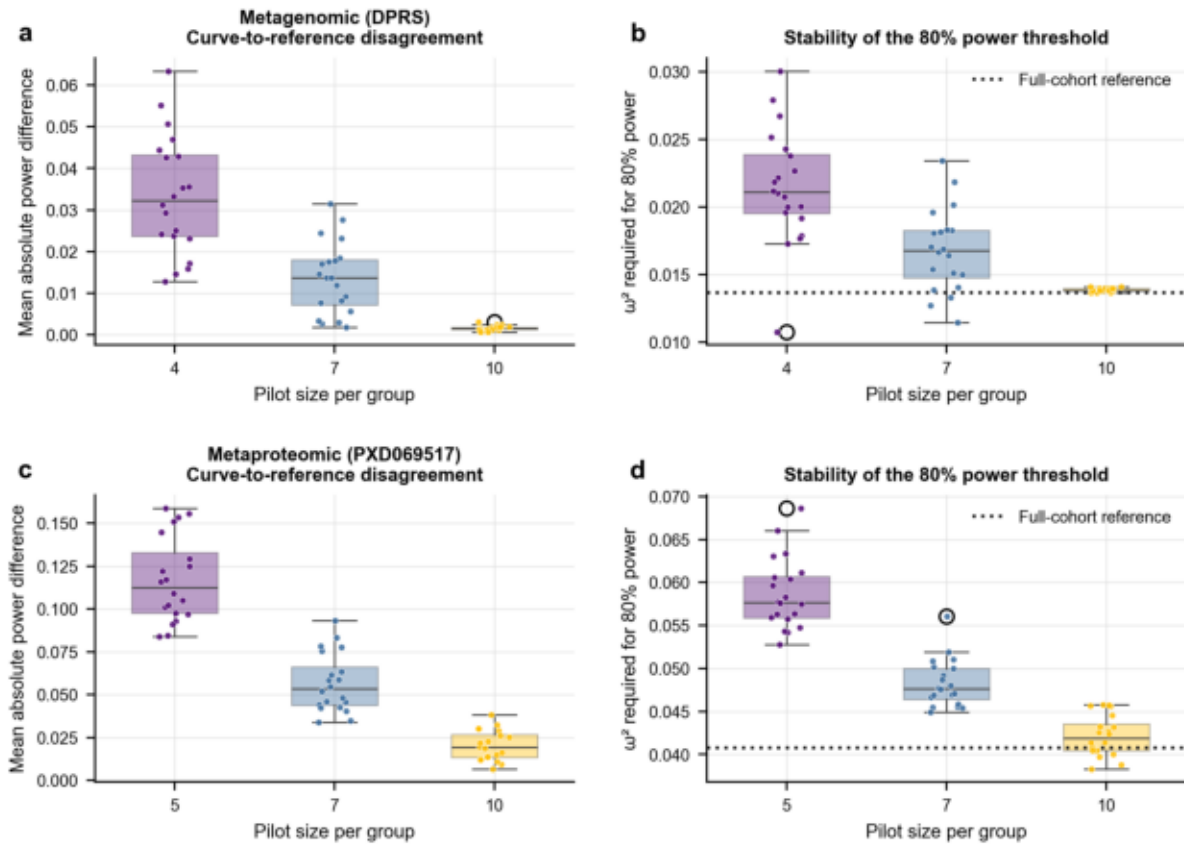
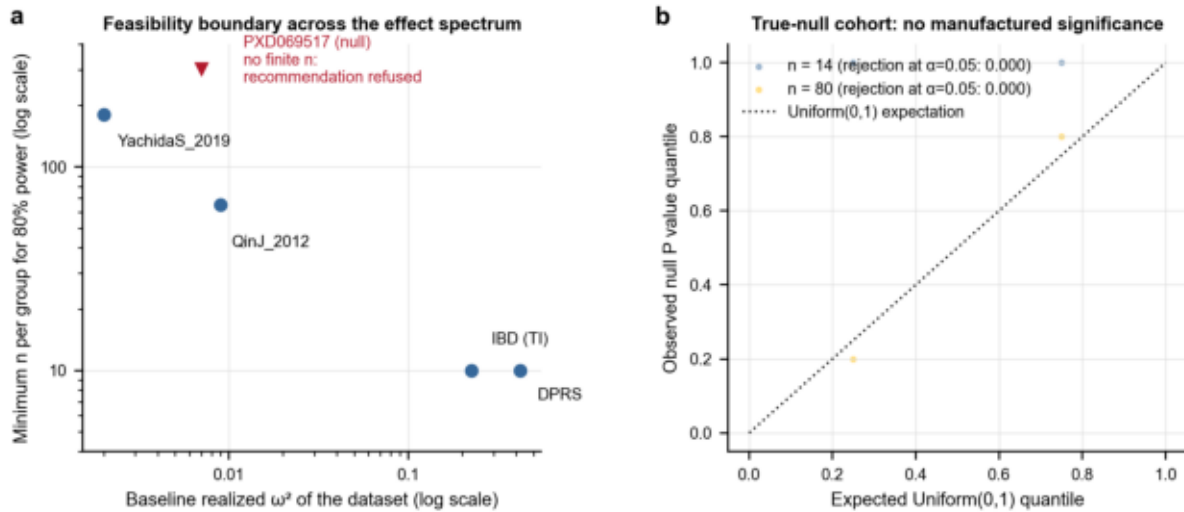


Supplementary Figures



Supplementary Figure S1. Finite-pilot disagreement and convergence of power-curve extrapolation for the metagenomic (a,b) and metaproteomic (c,d) workflows. Group-balanced pilots of 4, 7, and 10 samples per group (metagenomic DPRS cohort) and 5, 7, and 10 per group (metaproteomic PRIDE PXD069517 cohort, 14 per group in full) were repeatedly drawn from each empirical cohort and analyzed with identical effect grids, evaluation sizes, generator settings, and common random-number streams (20 repeated pilot draws per size). (a,c) Mean absolute power difference between each pilot-derived curve and the full-cohort reference curve decreases as pilot size increases; points represent repeated pilot draws and boxes summarize their distributions. (b,d) The realized ω^2 required to reach 80% power converges toward the full-cohort reference (dotted line) with increasing pilot size. The PXD069517 cohort shows no genuine between-group difference (realized $\omega^2 \approx -0.007$), so its panels validate the extrapolation mechanism only; its reference tree is a taxonomy-derived tree with unit branch lengths.



Supplementary Figure S2. Study-design feasibility across the effect-size spectrum and behaviour on a true-null cohort. (a) Minimum per-group sample size required for 80% power at each dataset's baseline realized ω^2 (log-log). Primary datasets (DPRS Cd vs Ni; pediatric IBD terminal-ileum CD vs Control) use PhyloPower power curves at the pilot-scale signal; because power rises from 0 to 1.0 between $n = 2$ and $n = 10$ on the archived grid, their boundaries are upper bounds. Independent cohorts (QinJ_2012, YachidaS_2019) use empirical full-cohort subsampling truth (500 without-replacement draws per size, PERMANOVA with 999 permutations, $\alpha = 0.05$, Bray-Curtis). The true-null PXD069517 cohort (realized $\omega^2 \approx -0.007$) yields no finite recommendation. (b) Null P values from 10 independent exchangeable-label null pools $\times$ 100 relabelings of PXD069517, evaluated at 14 and 80 per group (199 permutations), against the Uniform(0,1) expectation; rejection rates at $\alpha = 0.05$ were 0.041 and 0.052.

a vs control)

b

c(CRC vs control) d

Supplementary Figure S3. Validation on two independent metagenomic cohorts. (a,b) QinJ_2012 (type 2 diabetes versus control); (c,d) YachidaS_2019 (colorectal cancer versus control); 20 per group each. Power curves derived from pilots of 7, 10, and 20 per group, evaluated at the observed size (a,c; $n = 20$ per group) and extrapolated to $n = 80$ per group (b,d) at target $\omega^2 = 0.03$; the dotted line marks 0.8 power. Curves derived from different pilot sizes nearly overlap, and null-pool rejection rates remained close to the nominal level (0.035–0.105 at $\alpha = 0.05$).